Pablo Zavala
Agentic Technologies
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# AI Workforce Odyssey: Workforce Transitions Under AI Pressure

## Model Overview

This model represents 28 heterogeneous workers belonging to sectors with different susceptibility to automation, and shows how their outcomes vary depending on internal attributes and external policy conditions shaping their adaptation trajectories. The simulation is built in NetLogo, where each tick represents one month and the full horizon spans 200 months (roughly 17 years of labor-market adjustment).

The theoretical foundation draws on the task-based framework of Acemoglu and Restrepo (2018), which posits that AI displaces workers through the routine-task content of their jobs rather than through blanket labor substitution. Workers whose jobs are predominantly routine—cognitive or manual—face the highest displacement risk; those with creative, analytical, or hybrid profiles are more resilient. This task-composition heterogeneity is the central driver of differential outcomes in the model.

Two scenarios are compared using the same model structure and the same random seed (42424), so all differences in outcomes are attributable to scenario parameters rather than stochastic variation. Tech-Driven Automation applies high automation pressure (rate 0.80) with limited training support (5 seats, 8-month courses, no subsidy). Human-Centric Support uses more selective AI adoption (rate 0.55) with stronger worker support (12 seats, 5-month courses, 20% subsidy bonus). Workers are the only agents; firm strategy and government policy are represented as scenario-level settings to keep the model tractable and interpretable.

## Worker Design

Workers differ by task type, routine share (susceptibility to automation), and adaptability. They move across five states: employed, at-risk, in-training, unemployed, and re-employed. Displaced workers receive government transfers rather than fictitious labor income, and retraining builds human capital by raising adaptability, which facilitates a transition back to an employment state.

| Cohort | Count | Shape | Routine Share | Base Adaptability |
|---|---|---|---|---|
| **Routine-Cognitive** | 8 | Circle | 0.75 – 0.95 | 0.20 – 0.50 |
| **Routine-Manual** | 7 | Square | 0.60 – 0.80 | 0.30 – 0.60 |
| **Creative-Analytical** | 6 | Triangle | 0.15 – 0.35 | 0.50 – 0.80 |
| **Hybrid** | 7 | Star | 0.40 – 0.60 | 0.40 – 0.70 |

*Table 1. Worker cohort attributes.*

Key transition rules govern movement between states. An employed worker becomes at-risk when routine share × effective automation exceeds the disruption threshold (0.50). At-risk workers apply for training—admission depends on adaptability, peer influence, and scenario-level subsidy, but is capped by available seats. Workers at-risk for more than 10 months without a seat fall into unemployment, producing queue-driven job loss distinct from direct displacement. Training completion leads probabilistically to re-employment or unemployment; in either case, adaptability rises, representing genuine human-capital formation. Re-employed workers stabilize into full employment after 15 months. Unemployed workers can re-enter training or be directly hired, but face monthly skill scarring (adaptability decay) that makes recovery progressively harder.

Household budgets are tracked explicitly. Employed, at-risk, and re-employed workers earn labor income proportional to adaptability and task premium. Unemployed and in-training workers receive government transfers equal to 60% of their most recent labor income. Consumption is a fixed share of cash-on-hand, and the residual becomes savings, allowing the model to track asset depletion and consumption inequality as system-level outputs.

## Emergent Mechanism

The main emergent mechanism is the interaction of a limited training system with local coworker spillovers on the workplace grid. Training capacity is explicitly capped (5 seats under Tech-Driven, 12 under Human-Centric), but clustering and uneven recovery are not programmed as outcomes; they arise because adjacent coworkers affect one another's training and recovery chances.

What is programmed: each worker calculates a peer-boost from their four orthogonal neighbors on a fixed workplace grid. Neighbors who are in training or recently re-employed increase the focal worker's willingness to apply for training (+0.05 and +0.08 per neighbor, respectively), while unemployed neighbors suppress it (−0.03 per neighbor). The training capacity constraint is a hard global cap.

What emerges: spatial clusters of recovery or persistent unemployment on the workplace grid. When one worker begins training, their neighbors become more likely to apply. If admitted, they in turn boost their own neighbors, creating self-reinforcing local recovery waves—but only where early adopters happen to be placed. Under Tech-Driven, scarce seats mean fewer training examples are visible on the grid, so spillover cascades are weaker and patchier. Under Human-Centric, higher throughput produces denser recovery clusters and stronger positive feedback. The model tracks this emergence through two metrics: average supportive coworkers (local density of positive examples) and peer-boost standard deviation (how unevenly distributed peer influence is across the grid).

**Scenario Results**

Both scenarios were run with the same random seed (42424), so the initial worker attributes, workplace grid assignments, and stochastic draws are identical. All differences in the table below are therefore driven by the scenario parameters alone.

| Metric | Tech-Driven | Human-Centric |
|---|---|---|
| **Peak unemployment rate** | 14.3% | 3.6% |
| **Share ever unemployed** | 46.4% | 7.1% |
| **Avg unemployment time (exposed)** | 34.4 months | 3.0 months |
| **Burden Gini (exposed)** | 0.478 | 0.000 |
| **Avg liquid assets (final)** | 0.41 | 0.51 |
| **Avg supportive coworkers (final)** | 1.25 | 0.36 |
| **Peer-boost SD (final)** | 0.063 | 0.023 |

*Table 2. Scenario comparison (seed 42424).*

Tech-Driven produces a larger share of ever-unemployed workers (46.4% vs. 7.1%) and longer average unemployment spells among exposed workers (34.4 vs. 3.0 months). This is a direct consequence of the higher automation rate (0.80 vs. 0.55) pushing more workers past the disruption threshold, combined with a training system that cannot absorb the resulting demand. Tech-Driven also generates higher total output (79.5 vs. 55.0), revealing a core efficiency–equity tradeoff: faster automation raises aggregate production but concentrates adjustment costs on displaced workers.

The bottleneck dynamics differ sharply. In Tech-Driven, 5 training seats handling 8-month courses yield roughly 7–8 graduates per year. With 28 workers under heavy pressure, this creates persistent queues: workers at-risk for more than 10 months fall into unemployment, where skill scarring further reduces re-entry chances. In Human-Centric, 12 seats with 5-month courses yield roughly 29 graduates per year, which is enough to absorb most displaced workers before the timeout threshold, and the 20% subsidy bonus raises admission and completion probabilities.

Peer spillover effects amplify these differences. Under Human-Centric, higher training throughput means more workers are visibly training or re-employed on the workplace grid at any given time. Their neighbors receive stronger peer boosts, creating a positive feedback loop that accelerates recovery across local clusters. Under Tech-Driven, sparser training examples and more unemployed neighbors create negative local signals that suppress training applications, reinforcing the bottleneck from the demand side as well as the supply side.

The burden Gini among exposed workers diverges sharply (0.478 vs. 0.000): under Tech-Driven, some displaced workers bear far longer unemployment spells than others, while under Human-Centric the two affected workers experienced equal durations. The critical difference is on the extensive margin: 46.4% of workers experience unemployment under Tech-Driven versus 7.1% under Human-Centric. The primary inequality channel is thus how many workers are displaced at all, driven by the routine-share gradient across task types. The peer-boost standard deviation (0.063 vs. 0.023) confirms that peer influence is distributed more unevenly across the grid under Tech-Driven, where recovery pockets coexist with unemployment clusters. The peer-effect coefficients (+0.05, +0.08, −0.03) were calibrated so peer influence is visible but not dominant; the qualitative Tech-Driven–versus–Human-Centric ordering is robust to reasonable perturbations of these weights.

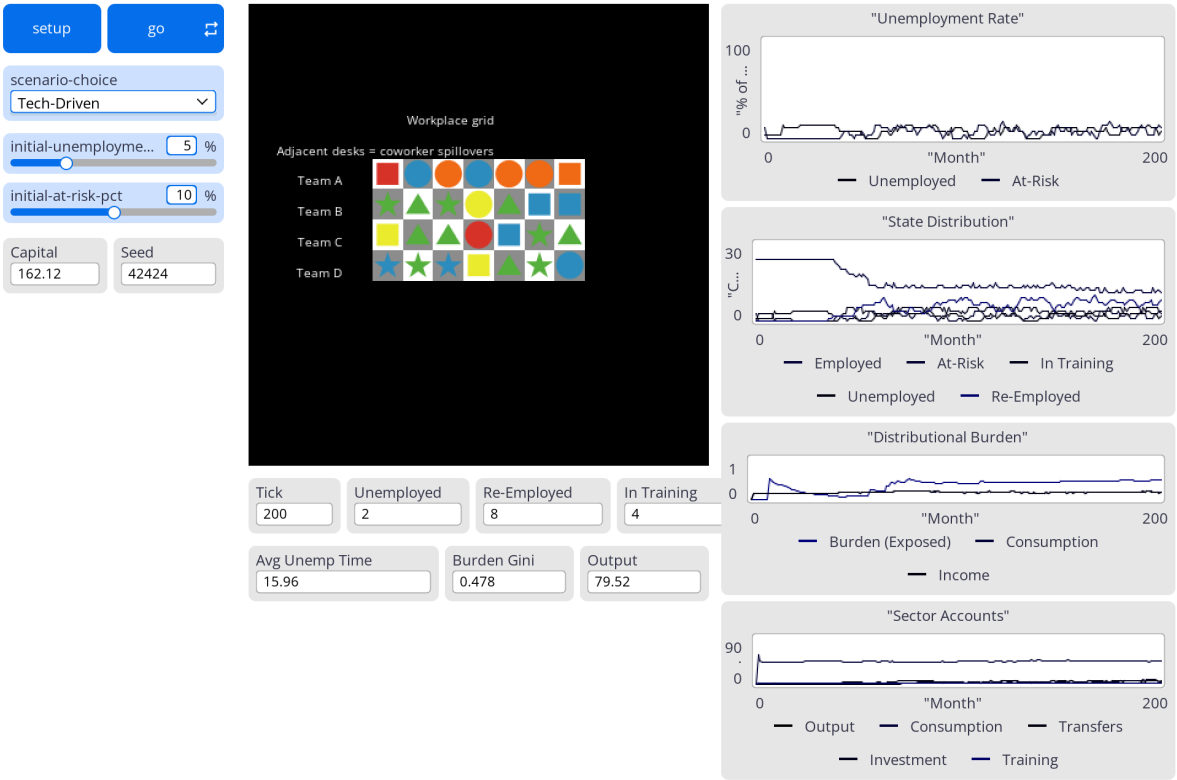


*Figure 1. Tech-Driven scenario at tick 200: workplace grid and system plots. Red agents indicate unemployed workers; yellow indicates at-risk; orange indicates in-training; green indicates employed; sky blue indicates re-employed.*

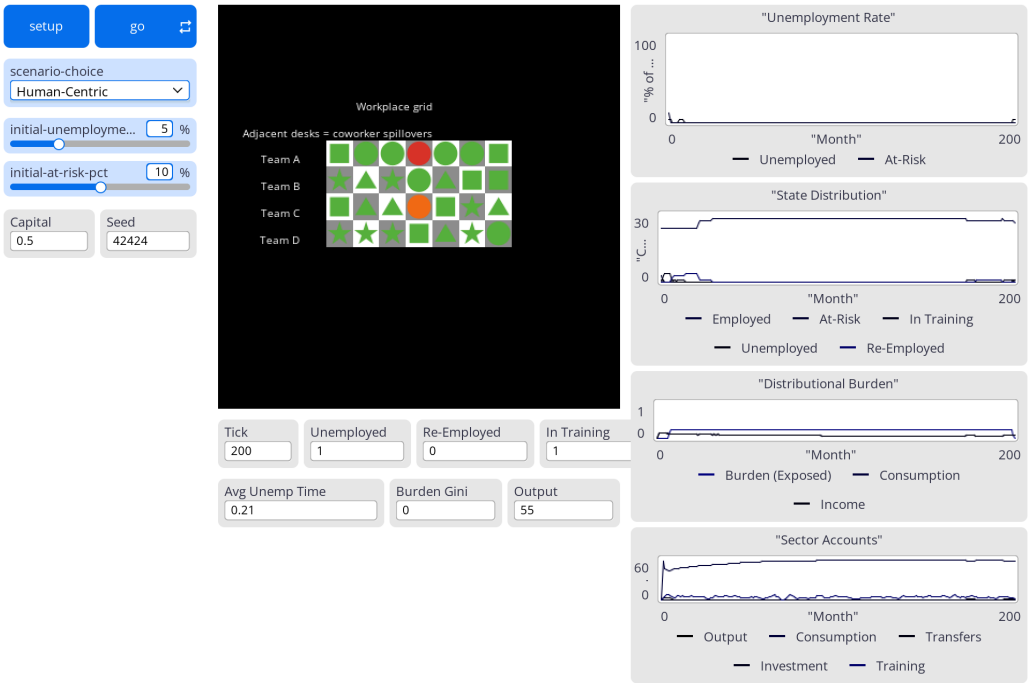


*Figure 2. Human-Centric scenario at tick 200: workplace grid and system plots. Note the higher density of green and sky-blue agents compared to Figure 1, reflecting lower displacement and stronger recovery.*

**Reflection**

Displacement thresholds, training admission rules, timed transitions, transfer support, and household-budget accounting are all explicitly programmed. Queue pressure, unequal recovery across worker types, and local recovery clusters are emergent. Instead, they arise from repeated local interactions among workers following simple rules, not from any instruction that commands clustering or inequality. The same programmed rules produce different spatial and distributional patterns across scenarios because the training bottleneck and peer-boost parameters interact nonlinearly with worker heterogeneity.

Several limitations should be noted. First, firms and government are not separate agents; the model cannot represent strategic firm responses or endogenous policy adjustment. Second, a single training pathway is available; real transitions involve heterogeneous programs with varying costs and durations. Third, workers do not relocate on the grid; in reality, job transitions reshape professional networks and peer-effect dynamics. Fourth, capital depreciation rises with installed stock to represent increasing maintenance costs, which produces natural convergence, but a full general-equilibrium closure with endogenous factor prices would yield richer dampening. Fifth, the goods-market gap is an accounting residual that diagnoses the distance from market clearing rather than a modeled equilibrium object. Despite these simplifications, the model demonstrates how modest policy differences, such as training capacity, course length, subsidy support, produce large aggregate differences through the amplifying effect of local spillovers and capacity constraints.

## References

Acemoglu, D., & Restrepo, P. (2018). The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment. *American Economic Review*, 108(6), 1488–1542.

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